



SDLC – Execution Roadmap & Debugging Prompts

Purpose: Follow this document step-by-step while building the project.

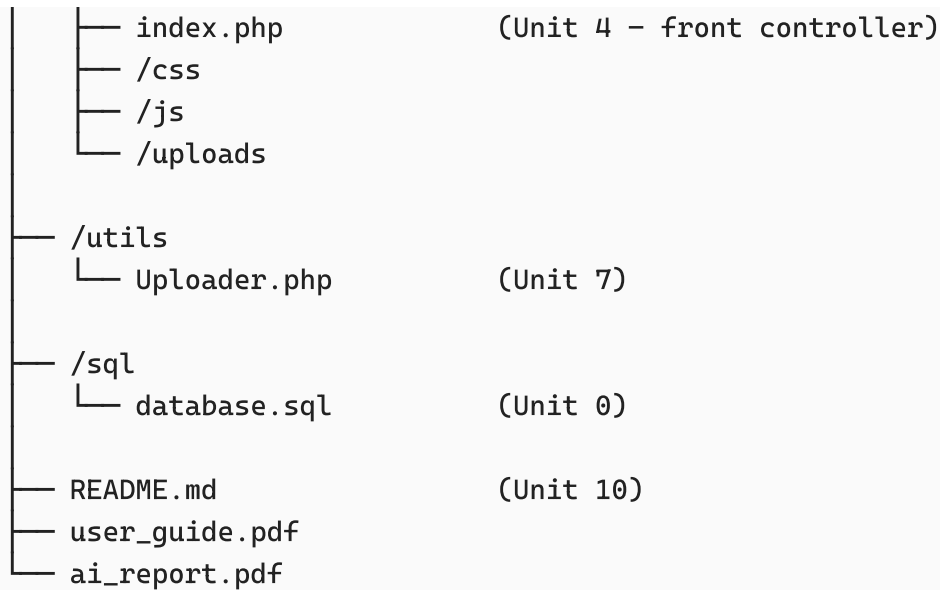
Each phase includes a **structural check** (what to verify) and **testing prompts** (copy-paste to AI to fix issues).



Project Structure (Must Remain Consistent)

Your root folder should look like this during execution:

```
project-root/
├── /config
│   └── Database.php           (Unit 1)
├── /models
│   ├── entities/
│   │   ├── User.php         (Unit 2)
│   │   ├── Product.php
│   │   └── Category.php
│   └── daos/
│       ├── UserDao.php      (Unit 3)
│       ├── ProductDao.php
│       └── CategoryDao.php
├── /controllers
│   ├── AuthController.php    (Unit 5)
│   ├── ProductController.php (Unit 6)
│   └── SearchController.php  (Unit 9)
├── /views
│   ├── /auth
│   │   ├── login.php
│   │   └── dashboard.php
│   ├── /products
│   │   ├── index.php
│   │   ├── show.php
│   │   ├── create.php
│   │   └── edit.php
│   └── /search
│       └── results.php
└── /public
```



⚠ **Never deviate from this structure** – vibe coding works only if every file knows where its neighbours live.

Execution Roadmap (6 Phases)

Phase 0 – Database & Environment (Day 1)

- ☐ Create MySQL database.
- ☐ Run `database.sql` from Unit 0.
- ☐ Create `config/db_config.php` (outside public) with credentials.

Test prompt (if database won't connect):

My PDO connection fails. Here is my code: [paste Database.php].
Error: [paste actual error].
Fix the connection using Singleton pattern. Ensure config file is included correctly.

Phase 1 – Core Infrastructure (Units 1–4)

Order:

1. Database singleton (`config/Database.php`)
2. Entity classes (`models/entities/*`)
3. DAO classes (`models/daos/*`) – test one (e.g., `UserDao::findAll()`)
4. Router + front controller (`public/index.php` , `Router.php` – can be inside `public/`)

Test prompt after integrating router:

I have a router in `public/index.php` that dispatches to controllers.
I created a `TestController` with an index action that echoes "hello".
When I visit `index.php?route=test/index` I get a 404.
Here is my `Router.php` code: [paste].
Fix the routing logic to parse the route parameter correctly.

Phase 2 – Authentication (Unit 5)

- ☐ Create `AuthController` with login/logout.
- ☐ Create login view.
- ☐ Protect any page: `Auth::requireLogin()`.

Test prompt (login not working):

My login system uses `password_verify()` but always returns false.
User password in DB is hashed with `password_hash()` during registration.
Here is my login method: [paste].
Error: [describe].
Fix the verification and session handling.

Error-fix prompt (session not persisting):

I call `session_start()` but `$_SESSION` is empty after redirect.
Here is my `AuthController` and the helper class.
Ensure session lifetime is correct and session ID regenerates after login.

Phase 3 – CRUD for One Entity (Unit 6)

- ☐ Generate `ProductController` + views.
- ☐ Test: list, create, edit, delete.
- ☐ Add CSRF token to forms.

Test prompt (create/store failing):

My create form POSTs to `ProductController@store`.
The DAO `save()` method returns true, but nothing appears in database.
Here are my controller and DAO code.
Find why insertion fails – check prepared statement binding and column names.

Error-fix prompt (delete doesn't work):

Delete button sends POST request, but record is not removed.
The DAO delete() method returns false.
Here is my delete() SQL and the controller.
Fix SQL error or foreign key constraint issue.

Phase 4 – File Upload & Admin (Units 7–8)

- ☐ Create `Uploader` class.
- ☐ Create `file_manager` table (SQL provided earlier).
- ☐ Build `FileManagerController` with upload/delete.

Test prompt (upload fails):

My `Uploader` class returns "invalid file type" for a .jpg file.
Allowed types include 'jpg'. The MIME type detection returns 'image/jpeg'.
Here is the `upload()` method.
Fix the extension/MIME validation logic.

Error-fix prompt (file not moved to target):

`Uploader` says success, but no file appears in `/public/uploads`.
Path is relative. Permissions are 755.
Check `move_uploaded_file()` error and ensure target directory exists.

Phase 5 – Search Module (Unit 9)

- ☐ Create `SearchController` + service.
- ☐ Build search form with filters.
- ☐ Paginate results.

Test prompt (search returns nothing):

My search query uses `LIKE %keyword%` but returns empty when I know matching rows exist.
Here is the `SearchService` code with the prepared statement.
Fix possible column name mismatch or SQL syntax error.

Error-fix prompt (pagination broken):

Pagination links show correct LIMIT offset, but the same records appear on every page.

Here is my SQL query with LIMIT :offset, :limit.

Fix the parameter binding – PDO treats placeholders as strings.

Phase 6 – Documentation & Integration (Unit 10)

- ☐ Run full integration test (create user, login, add product, upload file, search).
- ☐ Generate README.md (use Unit 10 prompt).
- ☐ Write AI report with the table from main project doc.

Final test prompt (after all units):

I have all units integrated. When I try to upload a product image and then search for that product, the image link is broken.

Paths are relative. The upload stores files in /public/uploads/products.

The product entity stores 'image_path' as 'uploads/products/filename.jpg'.

The view uses ``.

The page URL is /public/index.php?route=products/show/5.

Fix the asset path so the image loads.

Generic Error-Fixing Prompts (Use Anytime)

Problem	Prompt to Copy
White screen / no error	Enable full error reporting in PHP. Add <code>error_reporting(E_ALL); ini_set('display_errors', 1);</code> at the top of index.php. Then tell me what error appears.
PDO exception "column not found"	My SQL query uses column names that exist in the table. Here is the query and schema. Fix any typos or case mismatches.
Autoloader class not found	I'm not using an autoloader. I manually include files. Here is my index.php with <code>require_once</code> paths. Fix the paths so all classes load correctly.
Session data lost between pages	I call <code>session_start()</code> on every page. Session data disappears after redirect. Show me how to verify session save path and cookie parameters.
Form data not arriving in <code>\$_POST</code>	My form method is POST, but <code>var_dump(\$_POST)</code> is empty. The form HTML and controller are [paste]. Fix the form action or enctype.

Problem	Prompt to Copy
Foreign key constraint fails	I try to delete a user who has related products. Foreign key ON DELETE CASCADE should remove them but fails. Here is my table definition. Fix the constraint or delete logic.

Final Checklist – Project Ready to Submit

- ☐ All 10 units are completed and tested individually.
 - ☐ The router works for all controller actions.
 - ☐ Authentication protects pages correctly.
 - ☐ CRUD works without SQL errors.
 - ☐ Uploader stores files securely (unique names).
 - ☐ Search returns accurate results with pagination.
 - ☐ GitHub history shows regular commits.
 - ☐ README includes install steps + SQL dump.
 - ☐ AI report table is filled for every major AI assistance.
 - ☐ No [cite_start] or [cite: x] remain in final documents.
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